

Current Affairs 22-06-2026

Credit - Drishti Ias

Mombasa Declaration Adopted to Combat Illegal Fishing

Source: AP

The **Mombasa Declaration** was adopted at the **11th Our Ocean Conference in Mombasa, Kenya**, by countries from Africa, Asia, Europe, the Caribbean and the Pacific to strengthen action against **illegal, unreported and unregulated (IUU) fishing**.

- **Coalition for Fisheries Transparency:** It was developed with support from the **Coalition for Fisheries Transparency**, a global network of over 60 civil society organisations working to improve transparency and accountability in fisheries governance and management.
- **About the Declaration:** It is a call to action for **coastal and flag States to improve fisheries transparency**, especially through better collection, sharing and public access to information related to fishing vessels and fishing activities.
- **Governance Mechanism:** It supports the **Global Charter for Fisheries Transparency**, which outlines **10 low-cost policy principles** aimed at strengthening fisheries governance and ensuring sustainable marine resource management.
- **Key Commitments:** Signatory countries agreed to **introduce transparency reforms such as modernising vessel registries, publishing fishing authorisations, improving vessel ownership data** and strengthening information-sharing mechanisms.
- **Participating Countries:** Countries such as Belgium, Cameroon, Chile, Ghana, Guinea, Liberia, Panama, Papua New Guinea, Peru, Somalia, South Korea, France, The Gambia, Dominican Republic, Republic of the Congo and others endorsed the declaration, showing broad global support. (**India is not a signatory**).
- **Need:** IUU fishing threatens marine ecosystems, coastal communities, small-scale fishers, food security and livelihoods, especially in **developing and low-income countries** dependent on fisheries.
- **Impact:** IUU fishing is estimated to cost the global economy up to **\$50 billion annually, while also distorting markets**, reducing income for legitimate fishers and contributing to declining fish stocks.
- **Concerns:** The issue is linked to serious **human rights violations**, including **forced labour, exploitation, and unsafe working conditions** in parts of the global fishing industry.
- **Significance:** Countries that signed the Mombasa Declaration are expected to begin implementing its commitments, while a wider campaign will encourage more nations to join before the next **Our Ocean Conference in 2027**, making transparency a global norm in fisheries governance.

Read More: WTO Agreement on Fisheries Subsidies

Panchayats

For Prelims: 73rd Constitutional Amendment , Panchayati Raj Institutions , Model Youth Gram Sabha Initiative , People's Plan Campaign , Panchayat NIRNAY , 15th Finance Commission, State Election Commission (SEC) , SabhaSaar , SVAMITVA Scheme , eGramSwaraj , Rashtriya Gram Swaraj Abhiyan (RGSA) , Sustainable Development Goals (SDGs)

For Mains: Evolution and significance of Panchayati Raj Institutions in democratic decentralization, Role of Gram Panchayats in realizing Viksit Bharat and Localized SDGs (LSDGs).

Source: PIB

Why in News?

The Ministry of Panchayati Raj has released a review of the last 12 years of rural governance, highlighting significant progress in empowering **Panchayati Raj Institutions (PRIs)** through fiscal devolution, digital transformation, and grassroots capacity building in line with the vision of **Viksit Bharat** .

Summary

- Over the last 12 years, Panchayati Raj Institutions (PRIs) have been strengthened through enhanced fiscal devolution, digital governance platforms, AI-driven tools, capacity building, and greater participation of women, youth, and tribal communities.
- Despite these gains, challenges such as the 3Fs deficit (Functions, Funds, Functionaries), low Own Source Revenue, weak Gram Sabha participation, and inadequate administrative autonomy continue to hinder the realization of Viksit Bharat through grassroots governance.

What are the Major Reforms and Milestones in PRIs?

Fiscal Devolution and Autonomy

- **Finance Commission Grants:** There has been a transition in the quantum of tied and untied grants.
 - Rural Local Bodies received a record 94.98% of their allocation (Rs 2.82 lakh crore) under the **15 th Finance Commission (2020-26)** . The **16 th Finance Commission Projections (2026- 31)** has recommended an 84% increase, totaling Rs 4. 35 lakh crore.
- **Mobilizing Own Source Revenue (OSR):** To move away from complete grant dependency, the government introduced the **SAMARTH Panchayat Portal** and the **Atmanirbhar Panchayat Program** , equipping local bodies with end-to-end digital tools for revenue assessment, tax demand generation, and collection.

Capacity Building and Social Inclusion

- **Rashtriya Gram Swaraj Abhiyan (RGSA):** Revamped from 2022-23, this scheme has achieved

unprecedented institutional outreach. **RGSA** has completed the training of **4.10 crore Panchayat elected representatives and functionaries** across all three tiers.

- Between FY 2022-23 and FY 2025-26, over **33.55 lakh women elected representatives** were trained in governance, service delivery, and leadership.
- **PESA Act Fortification:** Rooted in the **Bhuria Committee (1995)** recommendations, the **Panchayats (Extension to Scheduled Areas) Act, 1996 (PESA)** empowers traditional Gram Sabhas to ensure self-governance in **Fifth Schedule Areas**.
 - To further institutionalize this grassroots democracy, the government has introduced a dedicated **PESA-GPDP Portal** that ensures local development plans strictly align with PESA guidelines and cultural sensitivities.
 - Additionally, India's first **PESA Ranking Framework** has been launched to foster competitive federalism by **evaluating states and districts on objective performance metrics**.
 - Indigenous-language training manuals have been introduced to overcome linguistic barriers and empower tribal representatives to effectively exercise their rights.
- **Women and Youth Leadership:** Initiatives such as the **Sashakt Panchayat-Netri Abhiyan**, which has trained over 33.55 lakh women representatives, and the **Model Youth Gram Sabha (MYGS)** align with the **National Education Policy, 2020** to nurture a strong pipeline of grassroots leadership.

Technological and AI-Driven Governance

- **SabhaSaar Platform:** Pioneering the use of **Artificial Intelligence**, this platform automatically generates **Gram Sabha Minutes of Meetings (MoMs)** in 23 Indian languages, bridging the linguistic divide and enhancing archival transparency.
- **Panchayat Advancement Index (PAI):** Introduced in April 2025, this data-analytics framework benchmarks Panchayat performance against the nine themes of **Localised Sustainable Development Goals (LSDGs)**.
- **Hyper-Local Forecasting:** Collaboration with the **Indian Meteorological Department** has led to India's first Gram Panchayat-level weather forecasting system, directly aiding agrarian decision-making.
- **Direct Citizen Participation:** The **Meri Panchayat App** has recorded more than **one crore downloads**, bridging the gap between rural citizens and local authorities by opening access to information regarding public assets and development projects.

Digital Governance and Algorithmic Administration

- **e-GramSwaraj and PFMS Integration:** **e-GramSwaraj portal** digitizes planning, budgeting, and accounting.
 - By mid-2026, over 2.59 lakh Gram Panchayats were onboarded. Its integration with the **Public Financial Management System (PFMS)** ensures real-time tracking, having processed over ₹3.16 lakh crore in digital transactions to eliminate fund diversion.
- **SabhaSaar:** To enforce transparency in local assemblies, this AI-powered tool utilizes the **BHASHINI platform** to automatically **transcribe and summarize Gram Sabha meetings** in 13 regional languages, significantly reducing proxy attendance and manipulated minute books.
- **Gram Manchitra and GPSDP:** A shift toward evidence-based spatial governance using **Geographic Information Systems (GIS)**. The Enhanced Gram Panchayat Spatial Development Plans (GPSDP) align local infrastructure with the **"One Nation One Map" framework**.

Historical Evolution

Ancient Roots: Panchayat-like systems existed since Vedic times, as referenced in ancient texts and historical records.

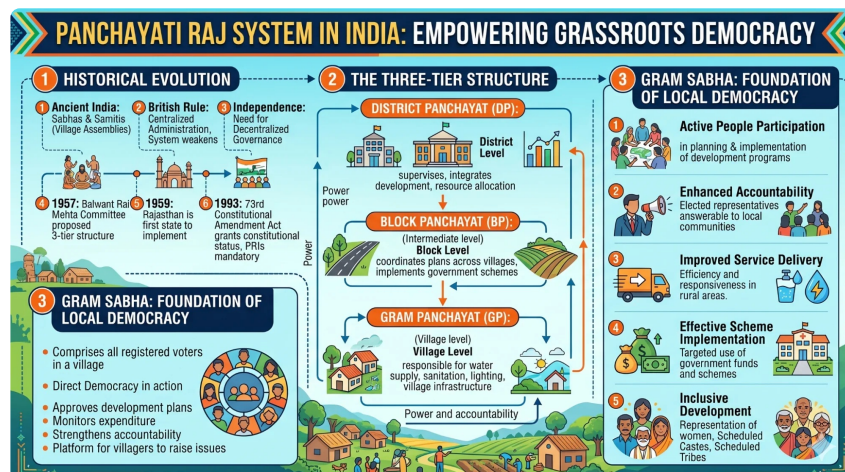
Colonial Era: British strategies (Mayo, 1870; Ripon, 1882) initiated decentralization. However, inadequate representation and powers limited their effectiveness.

Post-Independence: Article 40 of the Indian Constitution under Directive Principles mandated the creation and empowerment of village panchayats.

Constitutional Status and Structure

- **73rd & 74th Amendments (1992):** Conferred constitutional status to rural (Part IX) and urban (Part IXA) local bodies, mandating participatory local governance structures.
- **Current Status:** As of 2024–25, India has nearly 2,55,397 gram panchayats, 6,742 intermediary panchayats, and 665 district panchayats, underscoring the massive scale of grassroots democratic institutions.

#MainsWithDrishtiIAS



What are the Persistent Challenges Faced by PRIs in Realizing Viksit Bharat for Rural India?

- **Funds:** Heavy reliance on tied grants and state devolution. **Own Source Revenue (OSR)** generation remains abysmal, with PRIs depending on devolved funds for up to 95% of their revenues.
 - According to the **RBI's 2024 report on 'Finances of Panchayati Raj Institutions'**, PRIs generate a **mere 1.1% of their total revenue from local taxes and fees**.
- **Functions:** Incomplete devolution of the 29 subjects listed in the **11th Schedule**.
 - Data indicates that functional devolution has stagnated or declined in several states, limiting actual governance.
 - State governments frequently bypass PRIs by creating **Special Purpose Vehicles (SPVs), parastatals, or line-department committees** for executing schemes (e.g., rural infrastructure projects). This fragments local governance and erodes the constitutional authority of the Gram Panchayat.
 - Despite **Article 243E mandating timely Panchayat elections**, states often delay them citing delimitation, census, or administrative reasons. Courts frequently intervene to ensure elections are

held, as prolonged delays **weaken grassroots democracy and place panchayats under bureaucratic control. The '3Fs' Deficit (Functions, Funds, Functionaries):**

- **Functionaries:** A severe shortage of dedicated Panchayat cadres and technical staff (engineers, accountants), leading to excessive dependence on state-appointed bureaucrats and ad-hocism.
- **Structural and Social Barriers:**
 - **Proxy Representation:** The persistent phenomenon of '**Sarpanch Pati/Pradhan Pati**', where male relatives dominate the decision-making of elected women representatives, undermining the **33%+ reservation mandate**.
 - **Poor Gram Sabha Engagement:** National surveys indicate that Gram Sabha participation **remains critically low**, with entrenched caste divisions and gender biases limiting **meaningful citizen involvement in local governance**.
- **Irregular State Finance Commissions (SFCs):** Inordinate delays in the constitution of **SFCs** and the non-implementation of their recommendations severely cripple the financial autonomy and predictability of PRIs.

What Measures are Needed to Strengthen Panchayati Raj Institutions (PRIs)?

- **Operationalizing the '4Fs' Framework:** Ensuring the full, time-bound devolution of *Functions, Funds, Functionaries*, and **Freedom** to enable genuine local autonomy free from excessive bureaucratic control.
- **Institutionalizing Fiscal Empowerment:** Enhancing OSR mobilization by empowering Panchayats to levy local taxes and user charges.
 - Exploring modern financial avenues like the **monetization of rural assets and credit rating of high-performing PRIs**.
- **Granular Activity Mapping: Replace broad, vague delegations of the 11th Schedule's 29 subjects with strict, granular "Activity Mapping."**
 - Clearly delineate which specific tasks (e.g., primary school maintenance vs. teacher hiring) belong to the village, block, and district levels, eliminating overlap with state departments.
- **Panchayat Bonds:** High-performing PRIs should be credit-rated to access capital markets through local municipal/Panchayat bonds, drawing inspiration from **Indonesia's Dana Desa system**, which utilizes localized revenue from village-owned enterprises for self-sustenance.
- **Institutionalized Social Audits:** Following the **Mani Shankar Aiyar Committee(2012)** report, Gram Sabhas must be treated as supreme local deliberative assemblies.
 - Replicating **Meghalaya's Community Participation and Public Services Social Audit Act 2017**, independent social audit units and local Ombudsmen must be legislated nationwide to guarantee accountability.
- **Leveraging Emerging Technologies:** Promoting AI-based analytics for predictive planning, early warning systems, and asset management to make governance agile and future-ready.
- **Climate-Resilient Agriculture:** Interlocking local **Gram Panchayat-level weather forecasting** with agricultural advisory networks to insulate the rural economy from escalating climate vulnerabilities.

Conclusion

Realizing a *Viksit Bharat* requires transforming India's 2.59 lakh Gram Panchayats into micro-engines of economic growth and social equity. By matching the historic fiscal outlays with systemic

administrative freedom, professional cadres, and cutting-edge digital transparency, **India can shift from centralized governance to a highly efficient model of localized self-rule.**

Drishti Mains Question:

Q. Discuss the role of Panchayati Raj Institutions in achieving the vision of Viksit Bharat through decentralized governance and participatory development.

Frequently Asked Questions (FAQs)

1. What is the objective of the PESA Act, 1996?

The PESA Act extends self-governance to **Fifth Schedule Areas** by empowering Gram Sabhas to manage local resources, customs, and development planning.

2. What is the Panchayat Advancement Index (PAI)?

Introduced in **April 2025**, PAI is a performance benchmarking framework that evaluates Panchayats across the **nine themes of Localized Sustainable Development Goals (LSDGs)**.

3. What is the significance of e-GramSwaraj?

It digitizes Panchayat planning, budgeting, and accounting, while integration with **PFMS** ensures real-time financial monitoring and transparency.

4. What are the major challenges faced by PRIs?

Key challenges include the **3Fs deficit**, low Own Source Revenue (OSR), inadequate staff, weak Gram Sabha participation, and delays in State Finance Commission recommendations.

5. How does SabhaSaar improve local governance?

SabhaSaar uses **AI and BHASHINI** to automatically transcribe and summarize Gram Sabha proceedings in regional languages, enhancing transparency and record-keeping.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/aSf9m4lvH_E](https://www.youtube.com/embed/aSf9m4lvH_E)]

UPSC Civil Services Examination, Previous Year Questions (PYQ)

Prelims

Q1. Local self-government can be best explained as an exercise in (2017)

- (a) Federalism
- (b) Democratic decentralisation
- (c) Administrative delegation
- (d) Direct democracy

Ans: (b)

Q2. The fundamental object of the Panchayati Raj system is to ensure which among the following? (2015)

1. People's participation in development
2. Political accountability
3. Democratic decentralisation
4. Financial mobilisation

Select the correct answer using the code given below

- (a) 1, 2 and 3 only
- (b) 2 and 4 only
- (c) 1 and 3 only
- (d) 1, 2, 3 and 4

Ans: (c)

Mains

Q1. Assess the importance of the Panchayat system in India as a part of local government. Apart from government grants, what sources can the Panchayats look out for financing developmental projects? (2018)

Q2. To what extent, in your opinion, has the decentralisation of power in India changed the governance landscape at the grassroots? (2022)

Exercise Khaan Quest 2026

Source: AIR

An **Indian Army contingent** participated in the opening ceremony of **Multilateral Exercise Khaan Quest 2026 in Mongolia** , scheduled from 20th June to 3 July 2026.

- **About:** Exercise Khaan Quest is a multilateral military exercise that prepares participating defence forces for United Nations peacekeeping and peace support operations in a multinational environment.
- **Historical Background:** Started in **2003** as a bilateral exercise between the **United States** and Mongolian Defence Forces, it became a multilateral peacekeeping exercise in 2006, with the 2026 edition marking its 23rd iteration.
- **Participation:** The 2026 edition features **troops from 18 countries**, including a **40-member Indian Army** contingent comprising troops from a battalion of the JAT Regiment and personnel from other Arms and Services.
- **Objective:** The exercise aims to enhance interoperability, cooperation, operational readiness and coordination among participating militaries for peace support operations under **Chapter VII of the UN Charter**.
 - **Chapter VII of the United Nations Charter** deals with action that the **UN Security Council** may take in response to **threats to peace, breaches of peace, and acts of aggression** , including enforcement measures to maintain or restore international peace and security.
- **Training Focus:** Includes **joint planning, tactical drills, checkpoint establishment, cordon and search operations, patrolling, evacuation of civilians, counter-IED drills, combat first aid, and casualty evacuation** .
- **Strategic Significance:** India’s participation reflects its commitment to global peace, stability, international cooperation and UN peacekeeping, while also strengthening its strategic partnership with Mongolia.

Joint Military Exercises of India with Other Countries	
Name of Exercise	Country
Garuda Shakti	Indonesia
Ekuverin	Maldives
Hand-in-Hand	China
Bold Kurukshetra	Singapore
Mitra Shakti	Sri Lanka
Nomadic Elephant	Mongolia
Shakti	France
Surya Kiran	Nepal
Yudh Abhyas	USA

Read More: Exercise KHAAN QUEST

Sarvam Becomes India's Newest AI Unicorn

Source: FE

Bengaluru-based **startup Sarvam AI** has achieved **unicorn status** , becoming India's first dedicated full-stack **Artificial Intelligence (AI)** startup to be valued at over USD 1 billion. The milestone marks a significant boost for India's deep-tech and sovereign AI ecosystem.

- A "unicorn" refers to a privately held startup valued at over USD 1 billion. As of **June 2026**, **India has 131 unicorns**, making it the **world's third-largest unicorn ecosystem after the US and China** , with recent additions such as **Sarvam AI, Neysa, Skyroot, KreditBee, and Porter** highlighting a growing shift towards **deep-tech sectors, particularly AI and space technology** .
- **Sarvam AI** : It specializes in building India's **sovereign AI stack** . It focuses on reducing India's reliance on foreign AI platforms.
- **Indigenous Large Language Model (LLM) Mandate:** Under the aegis of the **Government of India's IndiaAI Mission** , Sarvam AI has been selected to develop the country's first indigenous LLM, constructing three distinct variants (**Sarvam-Large** for advanced reasoning, **Sarvam-Small** for real-time applications, and **Sarvam-Edge** for on-device operations) featuring up to **70-billion parameters** .
- **Unveiling of Frontier Models:** At the **India-AI Impact Summit 2026**, the startup unveiled two next-generation models **Sarvam-30B** (a 30-billion parameter model optimized for real-time conversations with a 32,000-token context window) and **Sarvam-105B** (a 105-billion parameter model with a 128,000-token context window tailored for complex multi-step reasoning).
- **Enterprise Platforms & Multilingual Tools:**
 - Sarvam Samvaad: Conversational AI agents that integrate with enterprise tools to generate insights and take actions using proprietary data.
 - Sarvam Dub: An AI dubbing model featuring zero-shot voice cloning and cross-lingual speech capability for seamless multilingual content creation.
 - Population-Scale Social Delivery: The startup introduced "**Vikram**" (a **multilingual chatbot named after Vikram Sarabhai**) and used its voice agents to collect high-quality data from 17 million farmers for the Ministry of Agriculture and Farmers Welfare.

Read more: Sarvam AI and the Sovereign AI in India

Supreme Court Declares Right to Walk a Fundamental Right

Source: TH

The Supreme Court of India delivered a landmark judgment in the case of ***Maniyar Iliyaz @ Shaik Riyaz v. P. Ayyappan & Ors. (2026)***, declaring that the **right to walk on safe and demarcated footpaths is a Fundamental Right**.

- **Surge in Pedestrian Fatalities:** Contextualizing the urgency, data reveals a catastrophic rise in pedestrian vulnerabilities. **Pedestrian deaths surged by nearly 163% between 2015 and 2024**, with their share in total road fatalities more than doubling from **9.5% to 20.61%**.
- **Constitutional Basis:** The **right to walk is an integral part of the right to free movement guaranteed under Article 19(1)(d)**, and it also embodies expressional, congregational, and associational rights under **Articles 19(1)(a), 19(1)(b), and 19(1)(c), read in conjunction with the right to life under Article 21**.
- **Absolute Priority over Vehicles:** The Court held that while the **freedom to walk is subject to reasonable restrictions**, pedestrians' right to use designated footpaths takes precedence over the movement and **privileges of motorized vehicles, and public spaces cannot be monopolized by drivers**.
- **Enforceable Correlative Duty:** The mere existence of a road creates an **enforceable legal duty on public bodies, specifically Urban Development Authorities, Municipal Corporations, Municipalities, and Panchayats**, to demarcate, construct, and maintain safe footpaths for walkers.
 - Citizens now have the right to invoke **restitutionary constitutional and legal remedies against these duty-bearers for failing to provide safe footpaths**, a mechanism that is **completely independent of the compensation remedies available under the Motor Vehicles Act, 1988**.
- **Critique of the Motor Vehicles Act, 1988:** The Apex Court criticised the existing legislation, noting that it **undermines walkers' rights by establishing the "vehicle" as the primary subject**, treating human and pedestrian interests as merely incidental.
 - Emphasising the need for modern institutional governance, the Court urged the formulation of dedicated laws and the establishment of an **independent, full-time regulatory body** with domain expertise to protect and plan for pedestrian rights.
- **Suo Motu Intervention:** The Court directed its Registry to forward the judgment to the **Law Commission of India** and key ministries including **Housing and Urban Affairs, Rural Development, and Road Transport and Highways** to initiate the necessary legal framework and identify duty-bearers.

Read more: Right to Safe Travel on National Highways under Article 21

Biochar and its Applications

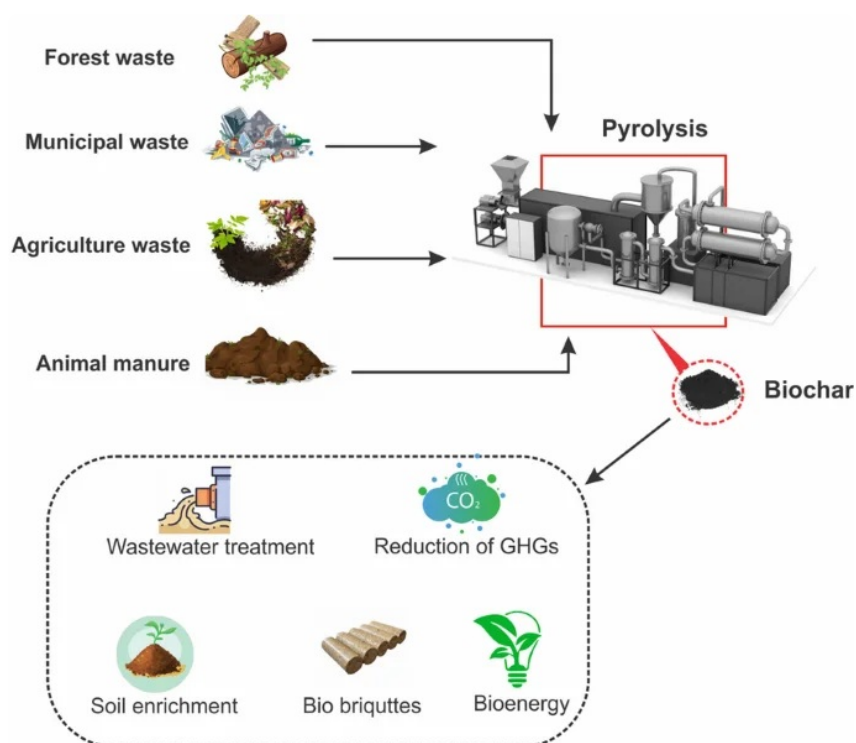
Source: TH

Why in News?

Biochar has emerged as a promising solution to two major challenges confronting Indian agriculture, namely large-scale crop residue burning and declining soil health. As India seeks climate-resilient and sustainable farming practices, biochar is gaining attention for its ability to **convert agricultural and organic waste into a valuable soil-enhancing, carbon-sequestering resource**.

What is Biochar?

- **Definition:** Biochar is a **highly porous, carbon-rich material** that resembles charcoal. It is produced through the thermal decomposition of organic biomass (such as agricultural waste, forestry residues, and food processing waste).
- **The Process (Pyrolysis):** It is generated through a process called **pyrolysis**, which involves heating the biomass in an oxygen-depleted environment at high temperatures (typically between **300°C and 700°C**).
- **Valuable By-products:** Along with solid biochar, the pyrolysis process also yields valuable by-products like **syngas** and **bio-oil**, which can be utilized as renewable energy sources.
- **Mechanism in Soil:** Its highly porous structure aggregates soil particles, effectively holding water and creating an ideal environment for beneficial microorganisms to thrive.



Significance and Applications

- **Enhancing Agricultural Productivity:** Applying biochar to degraded soils improves crop yields by **10% to 30%**, particularly in nutrient-deficient regions.

- It increases the soil's water-holding capacity by **10% to 25%** , helping crops withstand moisture stress during droughts and **heatwaves**.
- **Mitigating Air Pollution and Managing Waste:** India produces over **500 million tons** of crop waste annually. Converting **stubble into biochar provides a sustainable alternative to open-field burning in North India** , drastically reducing smoke and **particulate matter**.
 - It offers a solution for urban waste management by **converting biodegradable municipal solid garbage (which accounts for over 50% of India's 62 million tonnes of annual urban waste)** and sewage sludge, thereby diverting it from landfills.
- **Carbon Sequestration and Climate Action:** Biochar locks atmospheric carbon into a stable form that persists in the soil for centuries, preventing its release as **methane** or CO₂ .
 - One ton of biochar can sequester **2.5 to 3 tons of CO₂ equivalent** , making it a powerful nature-based climate solution.
- **Economic Viability via Carbon Credits:** Internationally recognized as a persistent carbon dioxide removal technology, each tonne of certified biochar can generate **2 to 2.8 tonnes of CO₂-equivalent in carbon credits** .
 - This creates a lucrative alternative income stream for rural communities, cooperatives, and farmers who can sell it as a soil amendment, biofuel, or trade the carbon credits.
- **Power Generation:** Biochar production through pyrolysis in India can generate **approximately 20-30 million tonnes of syngas and 24-40 million tonnes of bio-oil** as valuable byproducts.
 - Syngas can produce 8-13 Terawatt-hour (TWh) of electricity annually.
 - Bio-oil can substitute 12-19 million tonnes of diesel or kerosene, helping reduce crude oil imports and cutting fossil fuel emissions by over 2%.
- **Water Purification:** Chemically modified **pristine biochar** can be utilized in wastewater treatment to efficiently filter out toxic heavy metals such as chromium, arsenic, and lead.
- **Construction Sector:** Incorporating 2-5% biochar into concrete improves mechanical strength, increases heat resistance by 20%, and captures approximately 115 kg of CO₂ per cubic metre.

Domestic and Global Success Stories

- **India:** Field trials in the Akola district of Maharashtra using maize stalk biochar **improved overall soil fertility and organic carbon content in black soils**.
 - Research in Kerala showed that biochar derived from coconut leaf stalks significantly increased soil quality across different cropping systems.
 - The **KISAN kiln** project from IIT-Kharagpur is actively allowing smallholders to monetise their farm waste.
- **Global Examples:**
 - **Kenya:** Converting rice husks into biochar has produced thousands of certified carbon credits while improving soil phosphorus content and pH.
 - **Thailand:** The government has tied **biochar certification to its national carbon registry system** , driving use through soil rehabilitation initiatives.
 - **Brazil:** Reported large yield gains and high carbon retention using biochar generated from sugarcane bagasse.

Frequently Asked Questions (FAQs)

1. What is biochar and how is it produced?

Biochar is a porous, carbon-rich material produced by heating biomass in an oxygen-limited environment through pyrolysis, typically at 300°C–700°C.

2. How does biochar help address crop residue burning?

It converts agricultural residues into a valuable soil amendment, reducing open-field burning, air pollution, and particulate emissions.

3. Why is biochar considered a climate mitigation tool?

Biochar locks carbon in a stable form for centuries and can sequester about 2.5–3 tonnes of CO₂ equivalent per tonne produced.

4. How can biochar generate economic benefits for farmers?

Farmers can earn income through improved crop yields, sale of biochar products, and participation in carbon credit markets.

5. What are the major by-products of the pyrolysis process?

Pyrolysis produces syngas and bio-oil alongside biochar, both of which can be utilized as renewable energy sources.

6. What role can biochar play in urban waste management?

Biochar can be produced from biodegradable municipal waste and sewage sludge, reducing landfill burden and promoting waste-to-wealth initiatives.

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[Watch Video on YouTube: [▶ https://www.youtube.com/embed/owuX0EUq3go](https://www.youtube.com/embed/owuX0EUq3go)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Q. What is the use of biochar in farming? (2020)

1. Biochar can be used as a part of the growing medium in vertical farming.
2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for a longer time.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 2 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (d)

Q. With reference to the usefulness of the by-products of the sugar industry, which of the following statements is/are correct? (2013)

1. Bagasse can be used as biomass fuel for the generation of energy.
2. Molasses can be used as one of the feedstocks for the production of synthetic chemical fertilizers.
3. Molasses can be used for the production of ethanol.

Select the correct answer using the codes given below:

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (c)

Bimal N. Patel Elected as ITLOS Judge

Source: TH

Indian jurist **Prof. Bimal N. Patel** has been elected as a Judge of the **International Tribunal for the Law of the Sea (ITLOS)** for the 2026–2035 term, strengthening India's profile in international maritime law and multilateral legal institutions.

- Bimal N. Patel is a member of the **UN International Law Commission**, Vice-Chancellor and Professor of Public International Law at Rashtriya Raksha University, and has earlier served as Director of Gujarat National Law University and Member of the **21st Law Commission of India** .

International Tribunal for the Law of the Sea (ITLOS)

- **About:** ITLOS is an **independent judicial body established in 1996** under the framework of the **1982 United Nations Convention on the Law of the Sea (UNCLOS)** to adjudicate disputes relating to the interpretation and application of the Convention.
- **Headquarters:** The Tribunal is based in **Hamburg, Germany**, and functions as one of the key dispute-settlement mechanisms under UNCLOS.
- **Jurisdiction:** ITLOS deals with disputes related to **maritime boundary delimitation, navigation, conservation and management of marine resources, protection of the marine environment, marine scientific research**, and matters under other international agreements that confer jurisdiction on it.
- **Composition:** ITLOS consists of **21 independent members** , elected by secret ballot by States Parties to UNCLOS, selected for their integrity, fairness and recognised competence in the law of the sea.
- **India and UNCLOS:** India is a party to **UNCLOS since 1995** and supports rules-based maritime governance, peaceful settlement of ocean disputes and multilateral cooperation in the maritime domain.
 - **Dr Neeru Chadha was the first Indian woman judge of ITLOS and only the second woman judge in the Tribunal's history** , while Prof. Bimal N. Patel's election further strengthens India's judicial presence in the Tribunal.
- **Significance:** The election reflects India's growing role in international maritime law, ocean governance, UNCLOS-based dispute settlement, **blue economy** regulation, maritime security and protection of marine resources, making it important for global diplomacy and India's maritime interests.

Read More: UNCLOS